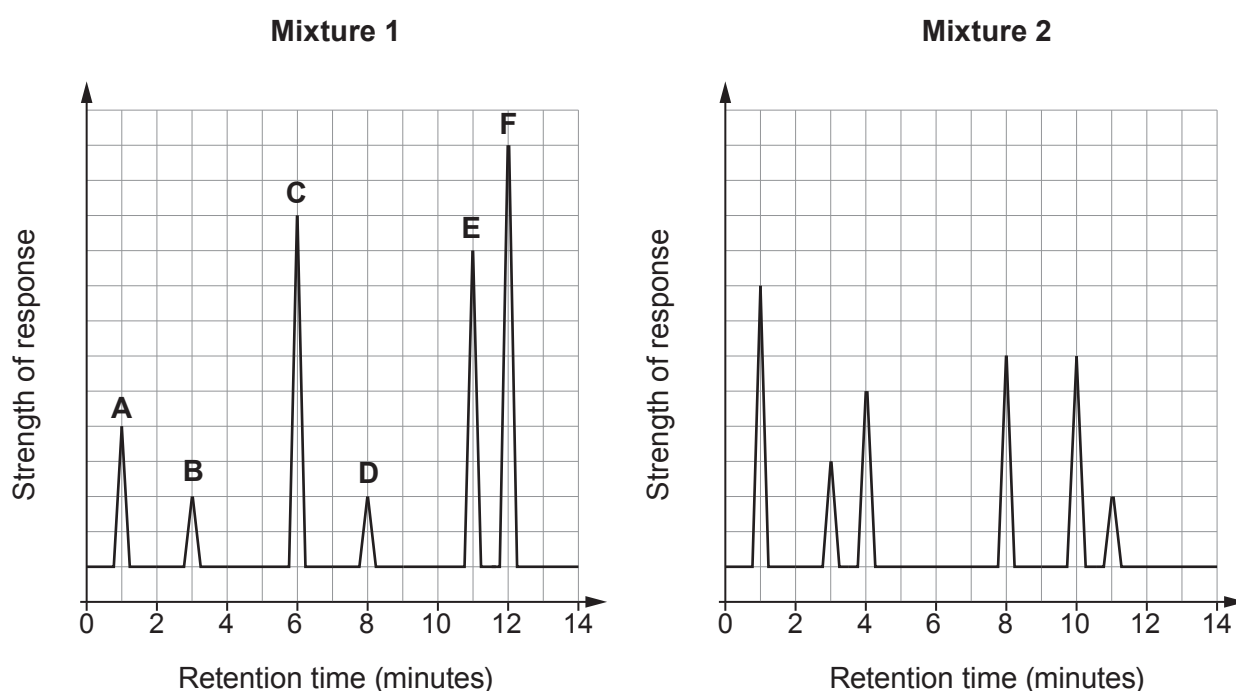


6. Chromatography is an important technique used by analytical chemists to separate substances found in mixtures.

Gas chromatography can be used to show:

- the number of compounds
- the identity of each compound
- how much of each compound is present in a mixture.

The diagrams below show the gas chromatograms from **mixtures 1** and **2**. Each mixture contains more than one compound. The compounds in **mixture 1** are labelled **A-F**.



- (a) Use the chromatograms to compare the similarities and differences between **mixtures 1** and **2**.

[4]

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- (b) Explain in terms of molecules why **F** has the greatest retention time.

[2]

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